

Fullstack Development

Stack Overflow 2024 Developer Survey

- Developer types
- Databases
- Web framework
- Tools

Fullstack Landscape

| Diagram

Inspired from this [VDO](#).

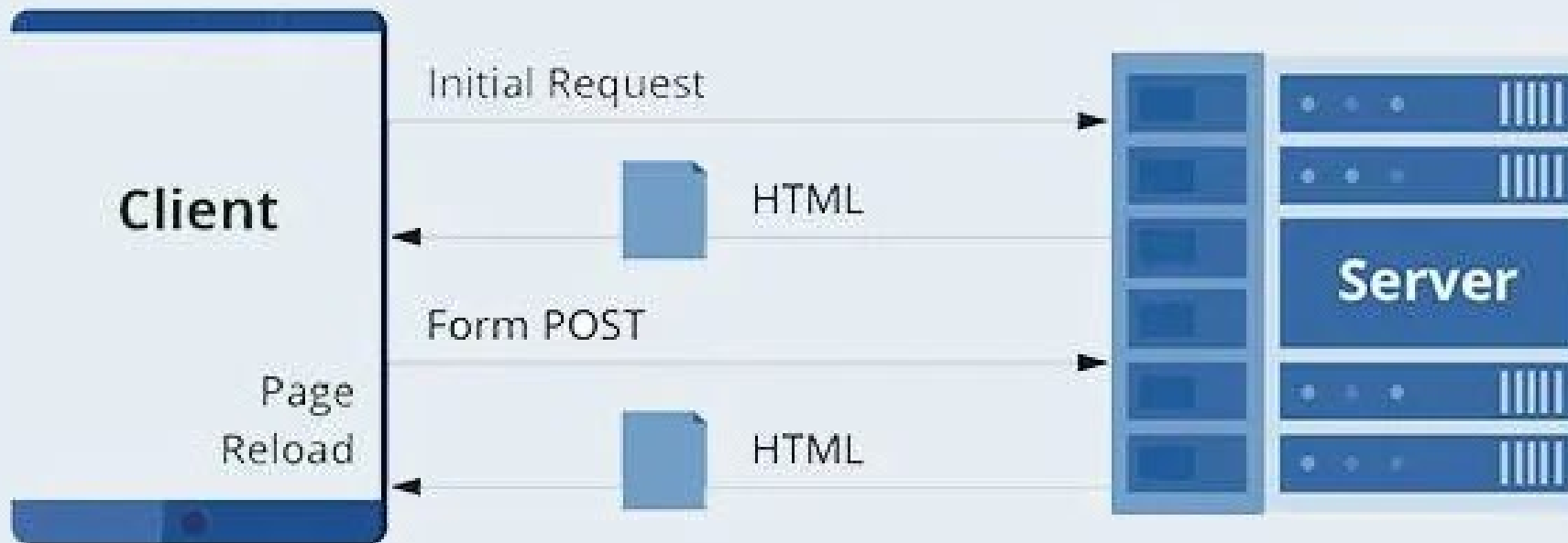
Part 1: The battle of the architectures

MPA vs SPA vs HTMX

Multi-page application

- Loads a new page every time you perform an action.
- Traditional web applications.
- Use server-side technologies
 - PHP, Ruby on Rails, ASP.NET, Java, and Node JS.
- Can include JavaScript (`script`) for client-side interactivity

MPA Lifecycle



Todo app (MPA)

- Express JS + Pug (renderer)

Get started

- `git clone https://github.com/fullstack-68/landscape-mpa.git mpa`
- `cd mpa`
- `pnpm install`
- `pnpm run build`
- `pnpm run start`

Endpoint

./src/index.ts

```
app.get("/", async (req, res) => {  
  const message = req.query?.message ?? "";  
  const todos = await getTodos();  
  // Output HTML  
  res.render("pages/index", {  
    todos,  
    message,  
    mode: "ADD",  
    curTodo: { id: "", todoText: "" },  
  });  
});
```

Renderer

```
./view/pages/index.pug
```

```
body
  main(class="container")
    a(href="/")
      h1 Todo (MPA)
    div(id="todoform")
      include ../components/inputform.pug
    div(id="todolist")
      include ../components/todolist.pug
```

The screenshot shows a web browser window with a single tab titled 'Todo'. The address bar shows 'localhost:3001'. The page content includes a title 'Todo (MPA)', a text input field, a 'Submit' button, and a list of todos. The first todo is '(1) 🍌 My First Todo' with delete and edit icons.

The browser's developer tools are open to the 'Network' tab. The 'Filter' dropdown is set to 'All'. The 'Blocked requests' and '3rd-party requests' checkboxes are unchecked. The network activity table shows three requests:

Name	Status	Type	Initiator	Size	T...
localhost	304	document	Other	943 B	5...
✓ pico.min.css	200	stylesheet	(index):0	(memory c...	0...
✓ pico.colors.min.css	200	stylesheet	(index):0	(memory c...	0...

At the bottom of the network panel, a summary bar shows: 3 requests | 943 B transferred | 158 kB resources | Finish: 537 ms | DOMContentLoaded: 538 ms | Load: 545 ms.

Use incognito mode to avoid loading chrome extensions.

Note

- Every button is wrapped in a separate form
 - Need to trigger different endpoints.
- Need to use `input(type="hidden")` to encode additional information.

```
form(action="/delete" method="POST" style="display: contents")  
  input(type="hidden" value={`${todo.id}`} name="curId")  
  button(type="submit" class="contrast" style="margin-bottom: 0") 🗑️  
form(action="/edit" method="POST" style="display: contents")  
  input(type="hidden" value={`${todo.id}`} name="curId")  
  button(type="submit" class="secondary" style="margin-bottom: 0") ✎
```

landscape-server

index.ts

```
src > index.ts > app.post("/create") callback
40
41 // Simulate latency
42 app.use(function (req, res, next) {
43   setTimeout(next, DB_LATENCY);
44 });
45
46 app.get("/", async (req, res) => {
47   // console.log(req.query);
48   const message = req.query?.message ?? "";
49   const todos = await getTodos();
50   res.render("pages/index", {
51     todos,
52     message,
53     mode: "ADD",
54     curTodo: { id: "", todoText: "" },
55   });
56 });
57
58 app.post("/create", async (req, res) => {
59   const todoText = req.body?.todoText ?? "";
60   try {
61     await createTodos(todoText);
62     res.redirect("/");
63   } catch (err) {
64     res.redirect(`/?message=${err}`);
65   }
66 });
67
68 app.post("/delete", async (req, res) => {
69   // console.log(req.body);
70   const id = req.body?.curId ?? "";
71   // console.log(id);
72 });
```

No type safety

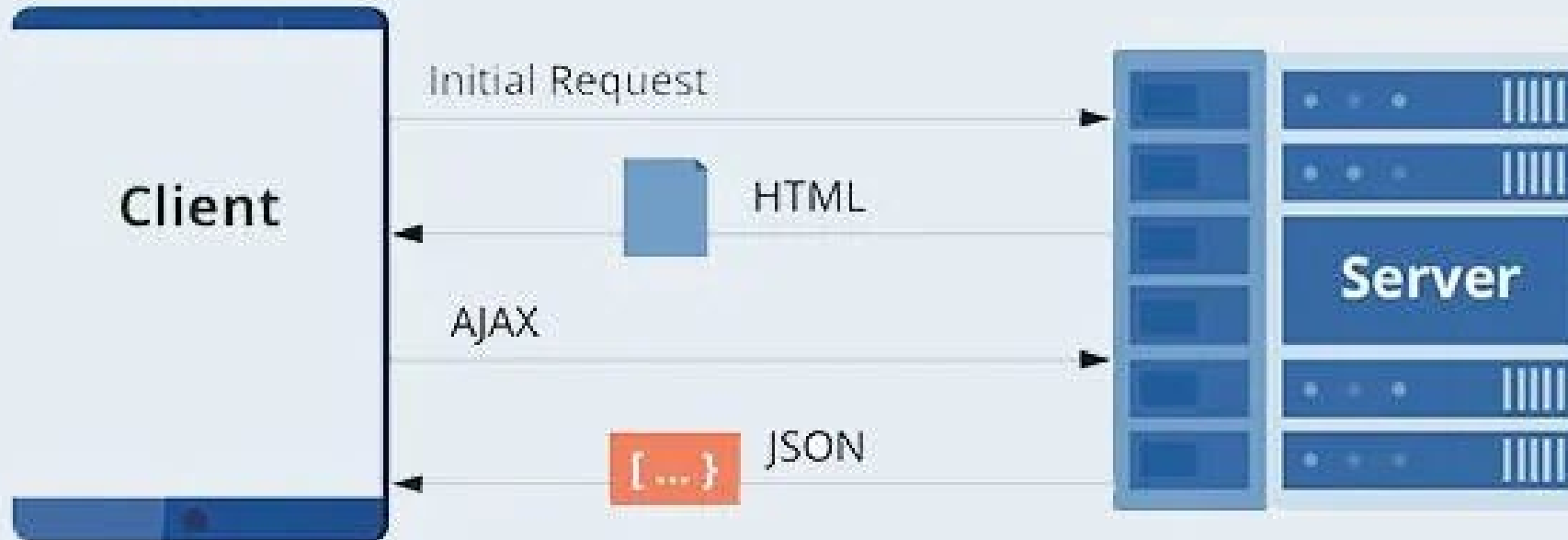
inputForm.pug

```
views > components > inputForm.pug
1 form
2 idTemplateColumns = mode === "ADD" ? "4fr 1fr" : "4fr 1fr 1fr"
3 ;="grid" style=`grid-template-columns: ${gridTemplateColumns}; align
4 (method="POST" action=`${mode=== "ADD"? "/create" : "/update"}` styl
5 input(type="text" name="todoText" value=curTodo.todoText)
6 button(type="submit")=`${mode=== "ADD"? "Submit" : "Update"}`
7 input(type="hidden" value=`${curTodo.id}` name="curId")
8 cancel button
9 mode=== "EDIT"
10 form(method="GET" action="/" style="display: contents")
11   button(type="submit" class="contrast") Cancel
12 ;
13 :class="pico-color-red-400")=message
```

Single-page application

- Single-page application
 - Loads a single HTML page and dynamically updates the content as the user interacts with the app.
- Use frontend and backend frameworks separately.

SPA Lifecycle



Todo app (SPA)

- Express JS + React

Get started

- `git clone https://github.com/fullstack-68/landscape-spa.git spa`
- Backend
 - `cd spa/backend`
 - `pnpm install`
 - `pnpm run build`
 - `pnpm run start`

Get started (cont.)

- Frontend

- `cd spa/frontend`
- `pnpm install`
- `pnpm run build`
- `pnpm run preview`

Vite + React + TS

localhost:5001

Todo (SPA)

Submit

(1) 🍌 My First Todo ✎ 🗑

Network

Filter

All Fetch/XHR Doc CSS JS Font Img Media Manifest WS Wasm Other

Blocked requests 3rd-party requests

100 ms 200 ms 300 ms 400 ms 500 ms 600 ms

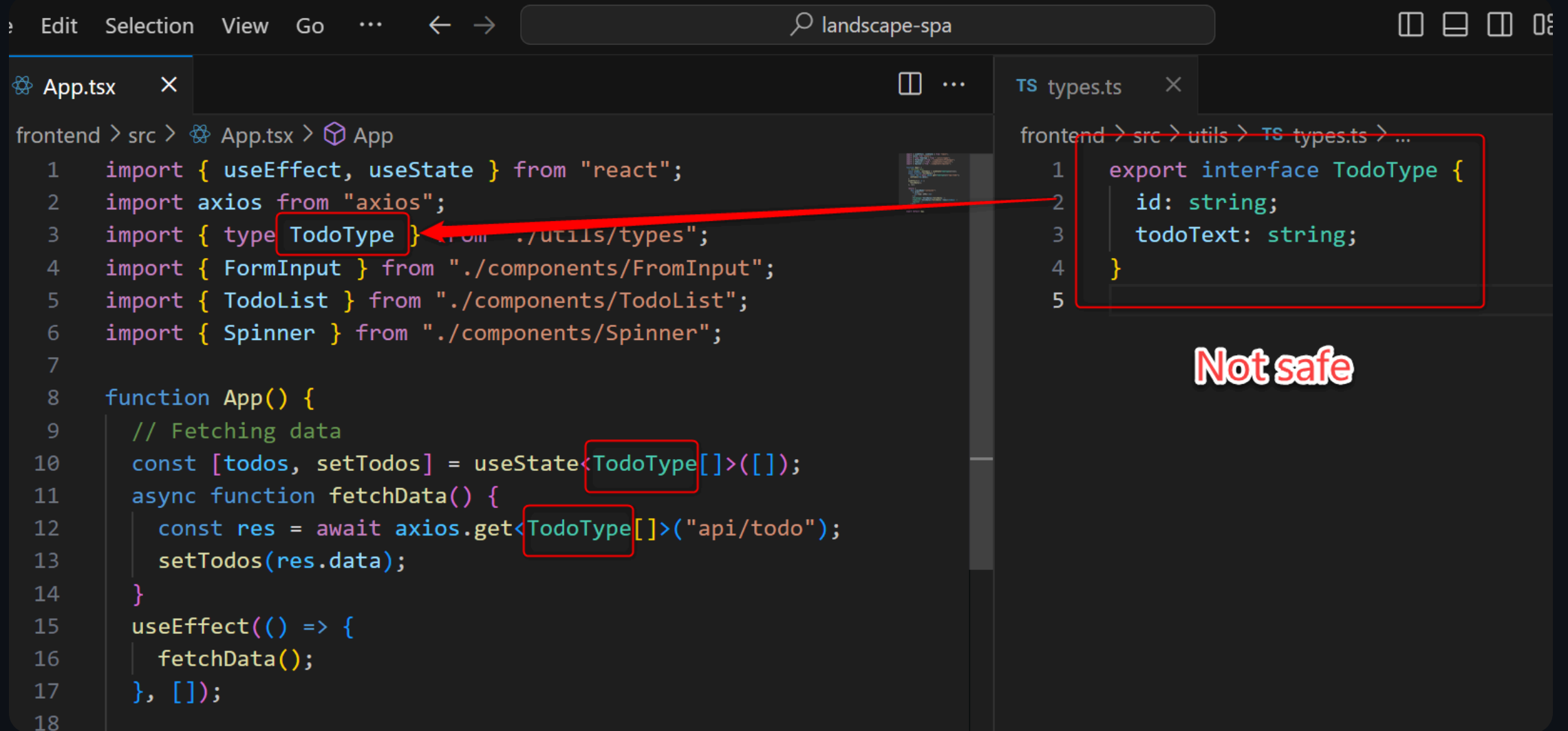
Name

- localhost
- index-uMjx6kOj.js
- index-z6Xg7xcP.css
- todo
- vite.svg

Headers Preview Response Initiator >>

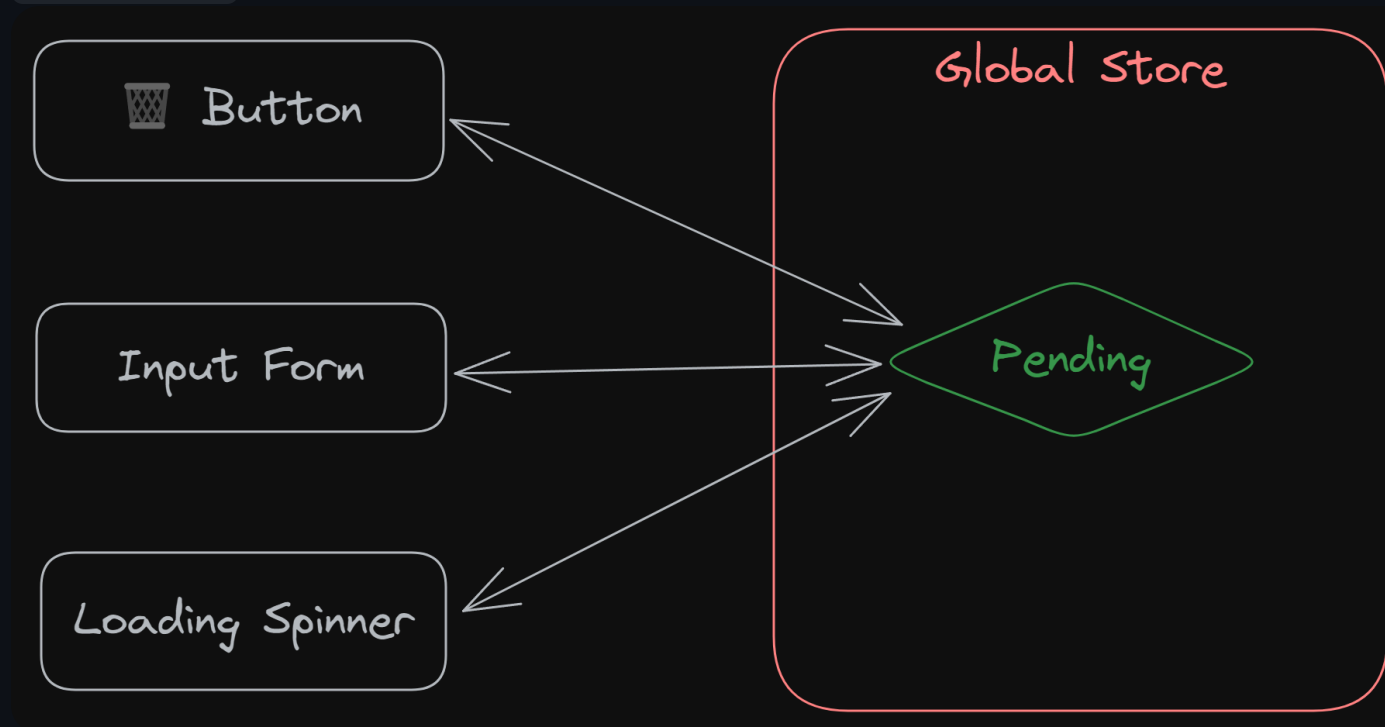
```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <link rel="icon" type="image/svg+xml"
6     <meta name="viewport" content="width=
7     <title>Vite + React + TS</title>
8     <script type="module" crossorigin src
9     <link rel="stylesheet" crossorigin hr
10  </head>
11  <body>
12    <main id="root"></main>
13  </body>
14 </html>
15
```

5 requests | 1.5 kB transferred | 343 kB resources | Finish: 5



Global store

- zustand



App comparison (UX)

Item	MPA	SPA
No page-refresh	✗	✓
Spinner	✗	✓
Element disabling	✗	✓

App comparison (technical)

Item	MPA	SPA
Amount of JS loaded	✓	✗
HTML content (SEO)	✓	✗
State in URL	✓	✗

DX

Item	MPA	SPA
Number of frameworks	1 	2 
Complexity	Less 	More 
Lines of code	Less 	More 
Type Safety	Less 	More 
Hot reloading	Partial 	Full 

Amount of Codes

Dir	# Files	Total Lines
<i>MPA</i>		
./src	2	161
./views	3	42
<i>SPA</i>		
./backend/src	2	144
./frontend/src	10	😭 360

Total: MPA=203, SPA=504

Round 3

Back to basic

HTMX

What exactly is HTMX?

Small JS library that can swap parts of UI with *HTML response* from a server.

2023 JavaScript Rising Stars

Get started

- `git clone https://github.com/fullstack-68/landscape-htmx.git htmx`
- `cd htmx`
- `pnpm install`
- `pnpm run build`
- `pnpm run start`

Todo app

- The stack is very similar to MPA (`express` + `pug`).
- SSR enabled (*duh!*)
- No reloading
- Spinner enabled
- Form input disabled during submission.

Code count

Type	# Line
MPA	203
SPA	504
RSC	292
RSC + RCC	527
HTMX	 259

Take that NextJS!

In addition

- Hypermedia as the Engine of Application State (HATEOAS)
- HTMX + Alpine.js

Summary

Architecture	Usage
MPA	Backend-critical (financial, ERP, ၈၃၂၀၆၃၃)
SPA	Dashboards, editor
RSC	Blogs, brochure sites
RSC + RCC	Highly interactive website with latest technology (to justify high price)
HTMX	Apps that finish on time